

MULTIOBJECTIVE URBAN PLANNING USING GENETIC ALGORITHM

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ABSTRACT: A genetic algorithm was used to search for optimal future land-use and transportation plans for a high-growth city. Millions of plans were considered. Constraints were imposed to ensure affordable housing for future residents. Objectives included the minimization of traffic congestion, the minimization of costs, and the minimization of change from the status quo. The genetic algorithm provides planners and decision makers with a set of optimal plans known as the Pareto set. The value of each plan in the Pareto set depends on the relative importance that decision makers place on the various objectives.

INTRODUCTION

The objectives of future planning for high-growth cities are many. One objective that the public is particularly sensitive to as a city experiences rapid growth is the minimization of traffic congestion. As more and more land is developed and sprawl becomes more entrenched, the public desire for the preservation of open spaces increases. Other objectives include:

- Control air pollution
- Provide affordable housing
- Maximize economic development
- Minimize taxes and fees
- Preserve historical and cultural sites
- Provide adequate utility infrastructure
- Minimize change
- Provide adequate education
- Provide adequate public safety

Many of these objectives compete with one another. It is impossible to develop plans that achieve all objectives. Trade-offs must be made based on the relative importance of one objective against another. One may think of these relative importances as weighting factors for the objectives. However, it is very difficult for planners to numerically quantify the relative weight that the public places on each of the objectives. The issue is further complicated by the dynamic nature of public sentiment. The most important objective today may not be the most important objective tomorrow. Without a clear statement of the relative importance of the competing objectives, can planners develop effective future plans?

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In this paper, the writers present a method for searching for a set of "Pareto-optimal" plans ("Pareto" plans for short). A plan is a member of the Pareto set if no other single plan has been found that is better in all objectives. A Pareto plan may be surpassed by another plan in one objective, and surpassed by yet another plan in another objective, but no single plan surpasses a Pareto plan in all objectives. The beauty of the Pareto set is that it is independent of the relative importance of the objectives. For a given combination of weighting factors between objectives, the optimum plan will be one member of the Pareto set. For another combination of weighting factors, the optimum plan will be another member of the Pareto set. In fact, the Pareto set includes optimum plans for all possible combinations of weighting factors. A non-Pareto plan is called a "dominated" plan, because a Pareto plan has been found that is better in every objective. The method described herein provides a great service to planners by eliminating the vast number of dominated plans from consideration.

With the Pareto set available, the city is still faced with the decision as to which plan to select from the set. This is where our work stops and the work of decision makers, policy makers, elected officials, and other stakeholders begins. The relative importance of objectives enters into this decision-making process, and the process can in no wise be automated. At least we have focused the attention of the decision makers on competitive Pareto plans. Thus, this paper presents an approach for performing the computational leg-work which makes the human, value-driven, decision-making process more effective.

The planning process is also complicated by the existence of an exponentially large number of possible options. A land-use plan for a city consists of assigning land uses to each of several zones throughout the city. Suppose a city has 200 land-use zones with 10 possible land uses for each zone. Then the number of possible land-use plans is 10^{200} . Because land use is coupled to other planning variables, the number of possible land-use plans must be multiplied by the number of possible street upgrade plans, transit plans, water/wastewater/stormwater plans, finance plans, school plans, etc., to get the total number of possible future plans for the city.

It is impossible for planners to search through and analyze all possible plans. In fact, it is even impossible to consider a significant fraction of the universe of possible plans. Oftentimes planners select a mere handful of candidate plans for analysis. This is unfortunate, because such plans are usually selected through an ad hoc process, and it is likely that much better plans exist. In this paper, the writers recognize that today's computing resources and planning models make it possible to consider many more plans. In the example presented in this paper, the writers consider millions of plans in our search for a Pareto set. We recognize, however, that this is still a very small fraction of the universe of possible plans. It is, therefore, unwise to generate millions of plans at random and take the Pareto set from that group. The Pareto set for a random group of plans may be far inferior to the Pareto set for the universe of possible plans. The writers use a genetic algorithm to search for a Pareto set that is much closer to the Pareto set of the universe of possible plans.

Genetic algorithms were first introduced by Holland (1975). They were later recognized to be general optimization techniques (Goldberg 1989). Genetic algorithms optimize by mimicking the evolutionary principles used by nature to optimize biological species. The application of genetic algorithms in design and planning has exploded in the past decade. For example, genetic algorithms have been used to optimize the design of:

- Helicopters (Crossley and Laananen 1996)
- Road construction excavation plans (Haidar and Naoum 1997)
- Building frames (Pezeshk et al. 1997)
- Sails (Day 1996)
- Tax structuring for multinationals (Fatouros et al. 1996)
- Floor plans (Rebaudengo and Reorda 1996)
- ATM networks (Cheng and Chang 1996)
- Bus route schedules (Thangiah and Nygard 1992)

The writers will first describe the example urban planning problem used in this study. Then we will explain how the genetic algorithm works. Finally, we will present results and conclusions.

EXAMPLE PROBLEM

The writers chose the city of Provo, Utah, as an example city for our study. Provo is a medium-sized city with a current population of about 107,000. It has experienced a steady growth rate of about 2.7% per year for the past six years. If this growth rate continues, the population of the city in the year 2020 will be about 192,000.

The writers divided the city into 130 land-use zones, as shown for a particular future plan in Fig. 1. A current GIS zoning map obtained from the city was used to derive these land-use zones. The goal was to produce a new land-use map of the city for the year 2020 that will accommodate a population of 192,000. The writers assigned a land-use variable to each of the 130 land-use zones. Each land-use variable can take on an integer value from 1 to 10. These values correspond to the ten land uses listed in Table 1.

The writers identified 25 major street corridors (existing and future) throughout the city, which are also shown in Fig. 1 for a particular plan. The capacities of the streets corresponding to these corridors were allowed to change. The capacities of all other existing streets were held fixed. The writers assigned a corridor variable to each of the 25 corridors. The possible integer values for each corridor variable correspond to the corridor types listed in Table 2. Note that increasing values correspond to increasing capacity. The writers only allowed upgrades in capacity from the status quo capacity of each corridor. Capacity downgrades were not allowed. An integer value of zero was only allowed for a corridor that has not yet been built.

In summary, we have 130 land-use variables and 25 corridor variables. A set of integer values for each of these variables constitutes a plan. Because there are ten possible integer values for each variable, there are 10^{155} plans in the universe of possible plans. We will now explain the objectives and constraints that were used to judge the quality of candidate future plans.

The writers required that six housing constraints must be satisfied. The first of these required that a future plan for the city must have enough housing to accommodate 192,000 residents. The housing capacity of a future plan is determined by summing over the residential land-use zones the product of the area and the housing density, and then multiplying the total by the current average number of residents per household. Then, we separated the current population of Provo into five income brackets and assumed that the population percentage in each bracket will be the same in the future population. The population percentages in these income brackets are shown in Table 3, along with housing preference fractions for each income bracket. These preference fractions were determined from census data indicating where people

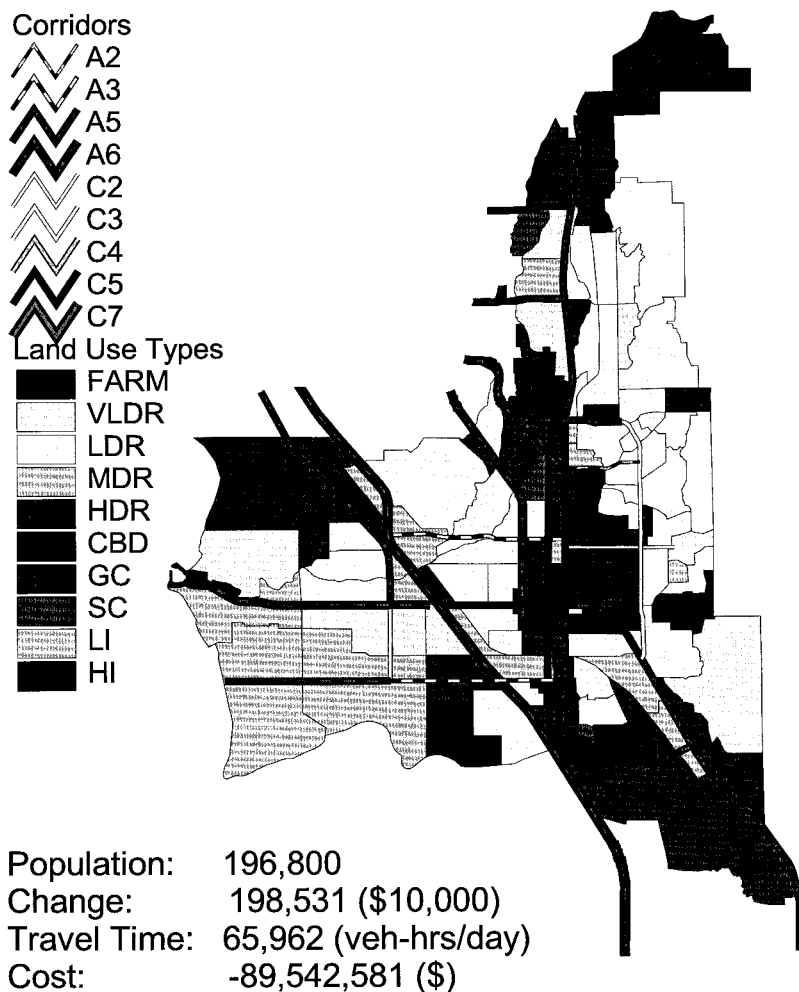


FIG. 1. Future Land-Use/Corridor-Upgrade Plan for Provo, Utah

currently live and survey data indicating where people would prefer to live if affordable housing were available. They represent a shift from the status quo to slightly higher percentages of higher density housing in the future. The writers imposed five constraints requiring that a future plan must provide at least 80% of the housing preferred by each income bracket.

The writers identified three objectives for our study. The first objective was the minimization of traffic congestion. We developed a traffic model for the city of Provo based on the MINUTP software package (*MINUTP* 1996). The traffic analysis zones for the model were identical to our land-use zones. The model takes the land uses for a future plan and estimates origin zones and destination zones for all trips throughout the city in a 24-hour period. Origins and destinations outside the city are included. The model routes all trips on streets throughout the city including the 25 major streets identified as corridors. The model reroutes trips as congestion and delays develop. The writers

TABLE 1. Possible Land Uses

Integer value (1)	Land use (2)	Description (3)	Housing density (homes/km ²) (4)
1	FARM	farm land	21
2	VLDR	very low density residential	371
3	LDR	low density residential	716
4	MDR	medium density residential	2,150
5	HDR	high density residential	4,990
6	CBD	central business district	—
7	SC	shopping center	—
8	GC	general commercial	—
9	LI	light industrial	—
10	HI	heavy industrial	—

TABLE 2. Possible Corridor Types

Integer value (1)	Corridor (2)	Description (3)	Average speed (km/hr) (4)	Capacity (veh/hr) (5)
0	C0	unused	—	—
1	C2	2-lane collector	53	920
2	C3	3-lane collector	56	1,380
3	A2	2-lane arterial	61	2,400
4	A3	3-lane arterial	64	2,600
5	C4	4-lane collector	60	3,160
6	C5	5-lane collector	64	5,200
7	A5	5-lane arterial	72	6,100
8	C7	7-lane collector	64	7,800
9	A6	6-lane arterial	72	9,150

TABLE 3. Housing Preference Fractions

Income bracket (1)	Population percentage (2)	FARM (3)	VLDR (4)	LDR (5)	MDR (6)	HDR (7)	Sum (8)
Lower	24	0	0	0.02	0.18	0.80	1.0
Lower middle	20	0	0	0.10	0.50	0.40	1.0
Middle	19	0	0.002	0.348	0.45	0.20	1.0
Upper middle	20	0	0.06	0.50	0.29	0.15	1.0
Upper	17	0.001	0.11	0.429	0.30	0.16	1.0

define the minimization of traffic congestion to be the minimization of the total travel time of all trips in a 24-hour period. We will refer to this objective as “travel time.”

Our second objective was the minimization of cost. The writers calculated the costs associated with the corridor upgrades required by future plans. These costs included construction costs and right-of-way purchase costs. We also recognized that changing the land use would affect the amount of property tax and sales tax revenues available to the city. The writers calculated the property tax and sales tax revenues for future plans. These calculations required that future property values be estimated for each land-use zone and future gross sales be estimated for each commercial zone. We neglected all

TABLE 4. Land-Use Degree-of-Change Factors

	FARM	VLDR	LDR	MDR	HDR	CBD	SC	GC	LI	HI
FARM	0	0.1	0.1	0.4	0.6	1.0	0.4	0.4	0.2	0.2
VLDR	0.3	0	0.5	0.6	0.8	1.0	0.5	0.6	0.7	1.0
LDR	0.4	0.3	0	0.2	0.4	0.7	0.5	0.6	0.7	0.8
MDR	0.6	0.5	0.4	0	0.1	0.6	0.4	0.5	0.8	0.9
HDR	0.9	0.8	0.6	0.3	0	0.4	0.6	0.6	0.7	0.9
CBD	1.0	1.0	0.8	0.6	0.2	0	0.2	0.2	0.4	0.9
SC	0.9	0.6	0.4	0.3	0.2	0.1	0	0.1	0.3	0.5
GC	0.9	0.6	0.5	0.4	0.2	0.1	0.1	0	0.1	0.5
LI	0.9	0.8	0.6	0.5	0.4	0.2	0.2	0.2	0	0.3
HI	0.9	0.9	0.8	0.6	0.4	0.5	0.5	0.5	0.3	0

other costs and revenues in the city, because we assumed that they would remain constant from future plan to future plan. Our "cost" objective was the minimization of corridor upgrade costs minus property and sales tax revenues until the year 2020. This value of cost may be negative if the revenues considered exceed the costs considered.

The writers found that it was necessary to add a third objective: the minimization of change. A future plan that satisfies the housing constraints and has low travel time and/or cost may be politically unacceptable because it requires too many zoning and corridor changes from the status quo. There is usually resistance to rezoning and corridor upgrades by the residents and businesses affected. The writers developed a crude metric of this resistance by summing over the land-use zones the product of the status quo land value of the zone and a degree-of-change factor, and summing over the corridors the product of the status quo land value of a strip on each side of the corridor and a degree-of-change factor. These two sums were added together to form our "change" objective. The degree-of-change factor matrices are shown in Tables 4 and 5. In these tables, one identifies the status quo land use or corridor type down the left side and the future land use or corridor type across the top. Note that the land-use degree-of-change table is not necessarily symmetric. These factors can be determined in part by reviewing the city's past history of accepted and rejected zoning changes. Nevertheless, judgment was used to set most of these factors. Note that only half of the corridor degree-of-change table is filled in. This is because corridors are only allowed to upgrade. The corridor degree-of-change factors are directly proportional to the change in capacity.

TABLE 5. Corridor Degree-of-Change Factors

	C0	C2	C3	A2	A3	C4	C5	A5	C7	A6
C0	0	0.10	0.15	0.26	0.28	0.35	0.57	0.67	0.85	1.0
C2		0	0.05	0.16	0.18	0.24	0.47	0.57	0.75	0.90
C3			0	0.11	0.13	0.19	0.42	0.52	0.70	0.85
A2				0	0.02	0.08	0.31	0.40	0.59	0.74
A3					0	0.06	0.28	0.38	0.57	0.72
C4						0	0.22	0.32	0.51	0.65
C5							0	0.10	0.28	0.43
A5								0	0.19	0.33
C7									0	0.15
A6										0

GENETIC ALGORITHM

The goal of our genetic algorithm was to search for a set of plans that satisfy the housing constraints and are Pareto optimal for our travel time, cost, and change objectives. Plans that satisfy the housing constraints are called "feasible" plans. The genetic algorithm began with a set of 100 feasible plans called the "starting generation." Plans in the starting generation were obtained by randomly changing the land-use variables and randomly upgrading the corridor variables from their status quo values. Plans randomly obtained in this way were checked for feasibility. If they did not satisfy the housing constraints, they were immediately discarded. It was necessary to randomly generate 1,978,980 plans to find 100 feasible plans for the starting generation. At this point, the values of the three objectives (travel time, cost, and change) were computed for each of the 100 feasible plans in the starting generation.

The genetic algorithm constructs a second generation of 100 feasible plans from the starting generation. It then constructs a third generation from the second generation and so forth until the final generation of 100 feasible plans is obtained. The writers let the genetic algorithm run for 100 generations. We found that the most computationally intensive part of our software was the calculation of the travel time objective via the MINUTP traffic analysis model. On a 200 MHz Pentium processor, the MINUTP model required about 30 seconds to analyze a single future plan. Evaluation of the housing constraints, the cost objective, and the change objective required less than a

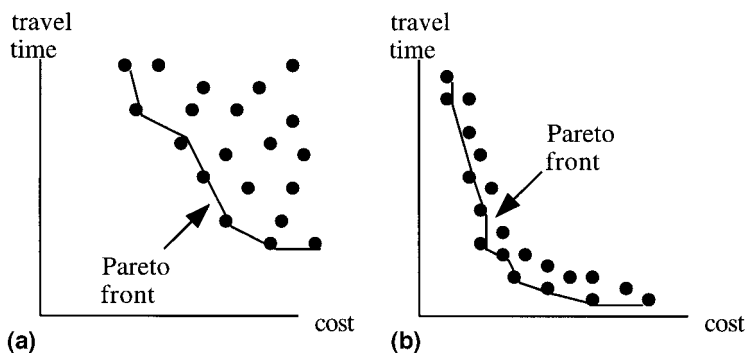


FIG. 2. Hypothetical Generations: (a) Starting Generation; (b) Final Generation

fraction of a second. Therefore, the writers decided to execute the MINUTP model only on feasible plans. Calculation of travel time for $100 \times 100 = 10,000$ feasible plans required about 3.5 days to execute.

Suppose that the writers had only two objectives (travel time and cost) rather than three, and that there were only 20 feasible plans in each generation rather than 100. Then we could plot travel time and cost on a graph for each of the feasible plans in the starting generation, as shown hypothetically in Fig. 2(a). We observe a fairly random scatter in values. Note that no single plan has both the lowest travel time and the lowest cost, which indicates that these objectives are competing. The Pareto set for the starting generation consists of the seven plans lying on the Pareto front. The Pareto set includes plans with low travel time, plans with low cost, and plans with low averages of travel time and cost. Depending on the relative importance placed on travel time and cost, decision makers would select one of the plans in the Pareto set as the best plan in the starting generation. Decision makers should not be interested in the dominated plans, because for every dominated plan, there exists a single Pareto plan that has both lower travel time and lower cost.

The genetic algorithm should produce a final generation of plans whose hypothetical plot is shown in Fig. 2(b). Note that 13 plans lie on the Pareto front in the final generation, whereas only seven plans lie on the Pareto front in the starting generation. Even the dominated plans lie close to the Pareto front. Furthermore, the Pareto front itself has shifted leftward and downward. We would hope that plans in the Pareto set of the final generation are close to plans in the Pareto set for the universe of possible plans.

In order for the genetic algorithm to achieve this hope, we need to devise a "fitness function," which is a measure of the Pareto optimality of each plan in a particular generation. Let cost_i , time_i , and change_i be the cost, travel time, and change, respectively, of the i th plan in the generation. Now consider two plans in the generation—the i th plan and the j th plan. The i th plan will be dominated by the j th plan if

$$\text{cost}_i > \text{cost}_j, \text{time}_i > \text{time}_j, \text{ and } \text{change}_i > \text{change}_j \quad (1)$$

This is equivalent to

$$\min(\text{cost}_i - \text{cost}_j, \text{time}_i - \text{time}_j, \text{change}_i - \text{change}_j) > 0 \quad (2)$$

Thus, the i th plan is a dominated plan if

$$\max_{j \neq i} (\min(\text{cost}_i - \text{cost}_j, \text{time}_i - \text{time}_j, \text{change}_i - \text{change}_j)) > 0 \quad (3)$$

The fitness of the i th plan is:

$$f_i = \left[1 - \max_{j \neq i} \left(\min \left(\frac{\text{cost}_i - \text{cost}_j}{\text{cost}_{\max} - \text{cost}_{\min}}, \frac{\text{time}_i - \text{time}_j}{\text{time}_{\max} - \text{time}_{\min}}, \frac{\text{change}_i - \text{change}_j}{\text{change}_{\max} - \text{change}_{\min}} \right) \right) \right]^p \quad (4)$$

In the above formula, the scaling factors $\text{cost}_{\max}$ and $\text{cost}_{\min}$ are the maximum and minimum costs, respectively, of all plans in the generation. The scaling factors for time and change are determined in similar fashion. According to (4), the fitness of Pareto plans will be between one and 2^p , whereas the fitness of dominated plans will be between zero and one. Since Pareto optimality is what we wish to achieve, the best plans will have the highest fitness values, and the worst plans will have the lowest fitness values. The exponent $p \geq 1$ in (4) will make the fitness of Pareto plans even higher and the fitness of dominated plans even lower. The writers used a relatively high value ($p = 15$), which makes the genetic algorithm quite aggressive in pursuing Pareto optimality.

Let us now explain how a succeeding generation is created from a preceding generation in the genetic algorithm. First, we copy ten of the best Pareto plans directly from the preceding generation to the succeeding generation. These plans are

1. The plan with the lowest travel time
2. The plan with the lowest cost
3. The plan with the lowest change
4. The plan with the lowest average travel time and cost
5. The plan with the lowest average travel time and change
6. The plan with the lowest average cost and change
7. The plan with the lowest average travel time, cost, and change
8. The plan with the highest fitness
9. The plan with the second highest fitness
10. The plan with the third highest fitness

There may be duplicates in the above ten plans—it doesn't matter. The remaining 90 feasible plans in the succeeding generation will be new plans. They are generated by repeatedly performing the following four-step process:

1. Selection
2. Crossover
3. Mutation
4. Abortion

In the selection step, two parent plans are selected from the preceding generation. This selection is random, but it is biased by the fitness function. The interval from zero to one on the number line is subdivided into 100 subintervals, corresponding to the 100 plans in the preceding generation. The length of each subinterval is proportional to the fitness of its corresponding plan. A random number is generated between zero and one. The plan corre-

sponding to the subinterval containing the random number is selected as the father. Another random number between zero and one is generated, and the plan corresponding to the subinterval containing the random number is selected as the mother. In this fashion, the most fit plans in the preceding generation are most likely to be selected as parents. Note that it is also possible that the same plan could be selected as both father and mother.

The father plan can be represented as a chromosome with 155 genes. The first 130 genes are the integer values (from one to ten) of the land-use variables in the plan. The last 25 genes are the integer values (from zero to nine) of the corridor variables in the plan. The mother plan can also be represented as a chromosome with 155 genes. The integer values of the genes of the mother chromosome will most likely be different from the integer values of the genes of the father chromosome. In the crossover step, we randomly generate an integer between 1 and 155 which defines the crossover gene. We cut the father and mother chromosomes right after the crossover gene and swap the portions of the chromosomes following the crossover gene. Thus, two new children chromosomes are created. The first child has identical values as the father chromosome in genes up to and including the crossover gene, and identical values as the mother chromosome in genes following the crossover gene. The second child has identical values as the mother chromosome in genes up to and including the crossover gene, and identical values as the father chromosome in genes following the crossover gene. In the rare case where the father and mother are identical, both children will be identical to the parents. Otherwise, the two children will represent two new plans which have characteristics of both the father and the mother. In this way the characteristics of the most fit plans in the preceding generation are incorporated into the new plans of the succeeding generation.

The mutation step allows completely new characteristics to occasionally creep into the generation. A random number between zero and one is generated for each gene in the two new children chromosomes. If the random number is less than the probability of mutation, the integer value of the gene is randomly changed to another value. Because mutations should rarely occur, the probability of mutation should be assigned a low value. The writers assumed a probability of mutation of 0.5%.

After the mutation step is completed for the two new children, the abortion step is executed. In this step the housing constraints are evaluated for each of the two children plans. If a plan violates any of the housing constraints, it is discarded. If it satisfies the housing constraints, it is added to the succeeding generation. The selection, crossover, mutation, and abortion steps are repeated until 100 feasible plans exist in the succeeding generation. On average, 29 plans were aborted in each generation. The travel time, cost, and change objectives as well as the fitness are calculated for each of the 100 feasible plans in the new generation.

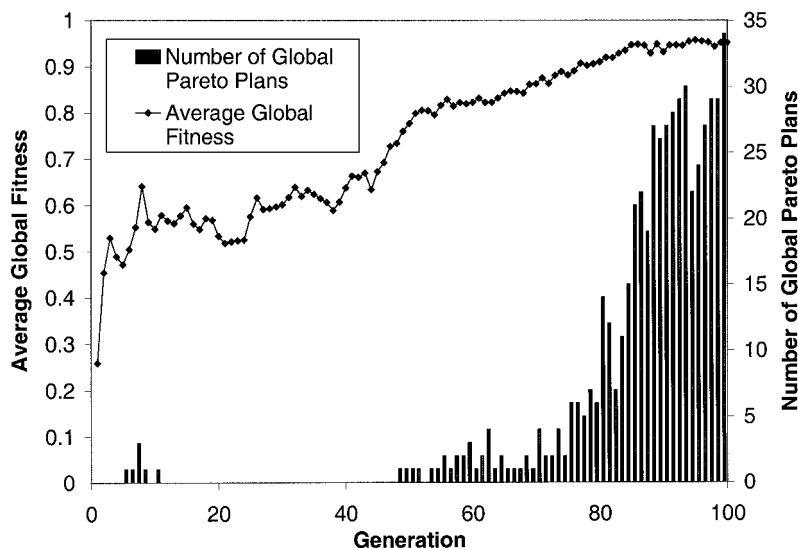
RESULTS

Table 6 gives generational averages and minima for raw values of the three objectives for the starting and final generations. Clearly, the genetic algorithm improved all three objectives. The Pareto set for the starting generation included 15 of the 100 feasible plans, and the Pareto set for the final generation included 52 of the 100 feasible plans.

The writers would like to examine how well the genetic algorithm improved Pareto optimality from generation to generation. However, our fitness function [(4)] was designed to compare the fitnesses between plans within a

TABLE 6. Results for Starting and Final Generations

Parameter (1)	Travel time (veh hours/day) (2)	Cost (dollars) (3)	Change (dollars) (4)
Starting generation average	159,900	-38,460,071	4,845,320,000
Starting generation minimum	36,282	-52,550,408	2,417,590,000
Final generation average	49,492	-62,960,606	2,196,880,000
Final generation minimum	31,012	-91,173,821	1,775,760,000

**FIG. 3. Improvement of Global Pareto Optimality from Generation to Generation**

generation, and one cannot compare fitnesses between plans in different generations. Therefore, the writers decided to create a new “global generation,” which included all 10,000 feasible plans from the 100 generations lumped together. We then calculated the “global fitness” for each of the 10,000 plans in the global generation according to (4). We also identified the “global Pareto set” for the global generation. Of the 10,000 plans in the global generation, there were 330 distinct plans in the global Pareto set. The writers averaged the value of global fitness over the 100 plans in each generation and plotted this average in Fig. 3. We also plotted the number of plans in each generation that are members of the global Pareto set. This plot clearly shows that the genetic algorithm improved global Pareto optimality.

In Fig. 4, the values of the three objectives for each of the 10,000 feasible plans in the global generation are plotted. This 3D plot has been projected on the plane perpendicular to the vector (1, 1, 1). It shows that plans in the global generation are well scattered over the three axes. However, there appears to be a bias towards plans with low values of change. This can be justified by the fact that these plans are feasible plans. The millions of infeasible plans that did not provide enough housing to accommodate the projected growth were aborted. It makes sense that feasible plans are biased towards a low degree of change from the status quo because the status quo

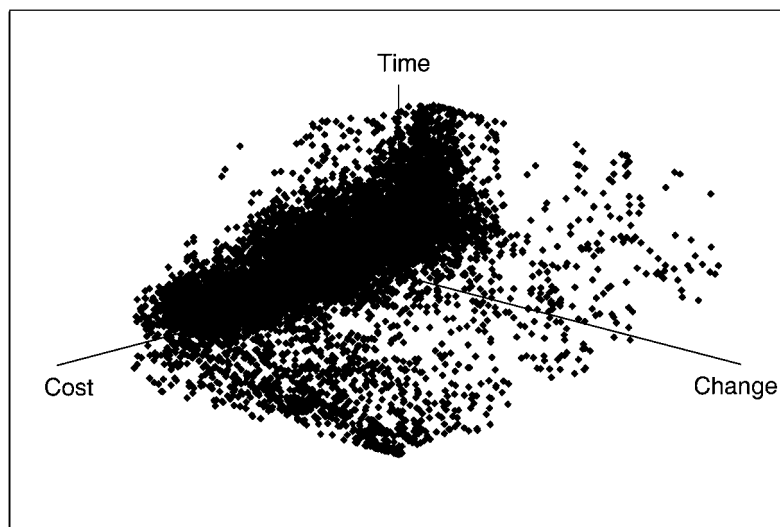


FIG. 4. Global Generation

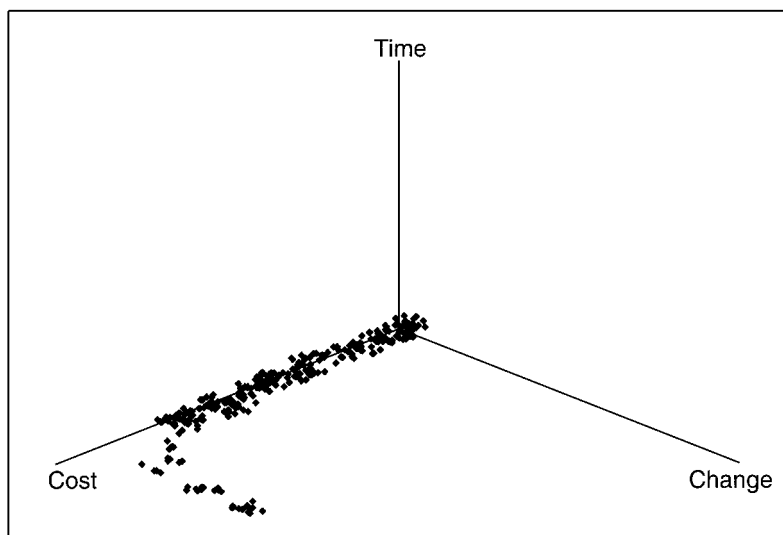


FIG. 5. Global Pareto Set

is itself a plan which has been “optimized” over the years to accommodate the growth that has occurred in the past.

Fig. 5 plots only the 330 plans in the global Pareto set. The difference between Figs. 4 and 5 dramatically illustrates the vast number of dominated plans in the global generation. Let us examine Fig. 5 further. The global Pareto set seems to form an L-shaped arc. Note that at one end of the arc are plans with low cost and change, while at the other end of the arc are plans with high cost and change. This indicates that the cost and change objectives are not strongly competing. This makes sense to some degree,

because low change plans would tend to have few corridor upgrade costs and retain the profitable sales and property tax revenue structure that has been developed over the years. Note that none of the plans has a very high value of travel time. The highest values of travel time occur at the extreme end of the arc where cost and change are lowest, and the lowest values of travel time occur at the other extreme of the arc where cost and change are highest. Thus, travel time competes with cost and change, although we do not see dramatically high values of travel time when cost and change are low. We observed that the highest values of travel time in the global generation were for plans whose land use radically differs from the status quo. This is because these plans use the existing network of corridors inefficiently. Because these plans also have a high change and cost, they are not members of the global Pareto set. Thus, the existing network or corridors used in our model is somewhat biased towards the land use of the status quo.

Fig. 1 shows the land use and corridor upgrades for one particular plan in the global Pareto set. This happens to be the plan which has the lowest average of the three objectives. If the three objectives have equal importance, this plan may be viewed as the best compromise plan. Similar land-use/corridor-upgrade maps can be created for each of the 330 plans in the global Pareto set. This set represents the best plans found by the genetic algorithm regardless of the weighting of the three objectives.

CONCLUSIONS AND RECOMMENDATIONS

A genetic algorithm with a unique fitness function was used to obtain a Pareto-optimal set of future land-use and corridor-upgrade plans for the city of Provo, Utah. The algorithm searched through 1,991,731 plans to get 10,000 feasible plans which provided enough housing to accommodate the projected future growth. From these 10,000 feasible plans, a global Pareto set of 330 plans was obtained which represents the algorithm's best plans for the three objectives (travel time, cost, and change) regardless of the relative importance of these objectives. The computational legwork required about 3.5 days.

The 330 plans in the global Pareto set must be narrowed by decision makers to a final plan or plans. This narrowing process will inherently involve the relative importances of the objective functions. At least two areas of future research are needed. First, a way to assimilate the overwhelming amount of detailed information associated with 330 plans is needed. The writers are currently working on a graphical tool that would display the land-use/corridor-upgrade map of a Pareto plan on the computer screen. In addition, slider bars would be available for each objective function. These slider bars represent relative weighting factors. As the decision maker moves the slider bars, the map displayed on the screen changes from Pareto plan to Pareto plan in real time. In this way the decision maker may rapidly observe planning trends from many Pareto plans. Second, formal procedures are needed to guide decision makers to the rational selection of a final plan. Such procedures should be rooted in decision theory.

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